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Role of HMOs in Preterm Nutrition

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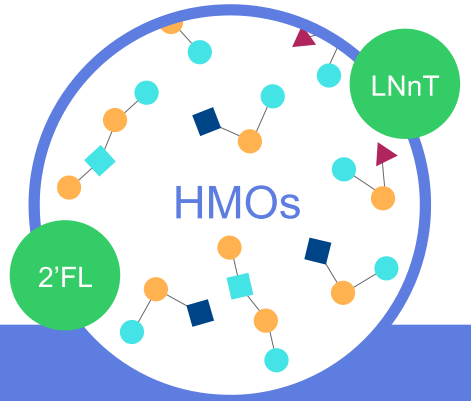

2. Role of HMOs in preterm infants

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Prematurity and Related Health Risks

Definition of Preterm Birth

Preterm birth: when a baby is born too early, before 37 weeks of pregnancy have been completed[1]

Terminology	Gestational age
Term	37 – 40 weeks
Preterm	< 37 weeks
Moderate / late preterm	32 - 37 weeks
Very preterm	28 - 32 weeks
Extremely preterm	< 28 weeks

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1. WHO 2018 (fact sheet preterm birth)

Preterm babies face specific health risks related to their immaturity / developmental needs[1-2]

- Respiratory Distress Syndrome
- Jaundice and higher risk for the brain
- Anaemia of prematurity
- Retinopathy of prematurity

- Feeding intolerance

- Feeding difficulties / metabolic disturbances
- Infections
- Brain injury
- NEC

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NEC, necrotising enterocolitis

1. WHO 2012; 2. WHO 2020 (Born too soon: the global action report on preterm birth)

An immature GI system leads to specific health risks in preterm infants[1-5]

Gastrointestinal system

Immature gut barrier

- Higher vulnerability to antigens and pathogens

Infection, allergy, NEC

Inefficient digestion and absorption

- Immature digestion of lactose, protein and fat
- Limited glucose absorption

Feeding difficulties, risk of insufficient nutrient intakes

Immature GI motor functions

- Poor suck-swallow-coordination
- Inefficient oesophageal motility
- Delayed gastric emptying
- Slow intestinal transit

Feeding intolerance

GI, gastrointestinal; NEC, necrotising enterocolitis.

1. Neu J, 2007; 2. Neu J, 2008; 3. Blackburn ST, 2013; 4. Omari TI and Rudolph CD, 2011; 5. Premji SS, 1998

A disturbed microbial colonisation of the GI tract contributes to feeding intolerance of preterm infants[1-8]

Disturbed gut microbial colonisation

Internal risk factors[7-10]

- Gestational age
- Birth weight
- Genetics
- In utero colonisation

- Delayed colonization of commensal bacteria, such as *Bifidobacteria*, *Lactobacillus*
- Limited microbial diversity
- Microbial disturbances persist up to 4 years of age

External risk factors[7-11]

- Antibiotics
- Mode of delivery
- NICU environment
- Skin to skin contact
- Breast milk exposure
- Medical complexity, inflammation, oxidative stress

Feeding intolerance, NEC, impact on immune system development and gut barrier integrity

GI, gastrointestinal; NEC, necrotising enterocolitis; NICU, neonatal intensive care unit.

1. Di Mauro A et al., 2013; 2. Ford SL et al., 2019; 3. Fanaro S, 2013; 4. Senterre T, 2014; 5. Fouhy F et al., 2019; 6. Yuan Z et al., 2019; 7. Collado C et al., 2015; 8. Hendrickx JGE et al., 2019; 9. Lenfestey MW and Neu J, 2018; 10. Unger S et al., 2015; 11. Greenwood C et al., 2014.

Up to 50% of preterm babies suffer from feeding intolerance[1]

Risk factors[2,3]

- Functional immaturity of the GI tract
- Lack of coordinated GI motility
- Impaired digestive function
- Disturbed gut microbial colonisation

Feeding Intolerance*

Consequences[2,3]

- Suboptimal nutrient intake
- Decreased growth rate
- Delayed achievement of full enteral feeding
- Prolonged dependence on intravenous nutrition
- Inability to meet criteria for NICU discharge

*Feeding intolerance can be defined by the inability to digest enteral feedings (GRV > 50%), abdominal distension or emesis or both, and the disruption of the patient's feeding plan.[2]

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GI, gastrointestinal; GRV, gastric residual volume; NICU, neonatal intensive care unit.

1. Indrio F et al., 2011; 2. Fanaro S, 2013; 3. Senterre T, 2014.

Achieving timely enteral feeding is critical to support and development of preterm babies

Benefits of timely enteral feeding[1-4]

- Promote growth
- Support normal development of the GI tract
- Promote nutrient intake
- Support neurodevelopment and long-term health outcomes

- Minimise the risk of NEC
- Avoid health problems related to feeding by intravenous lines

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GI, gastrointestinal; NEC, necrotising enterocolitis.

1. Fanaro S, 2013; 2. Senterre T, 2014; 3. Koletzko B, 2014; 4. Chen CM, 2012.

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Role of HMO in Preterm Infants

HMO, Human milk oligosaccharide.

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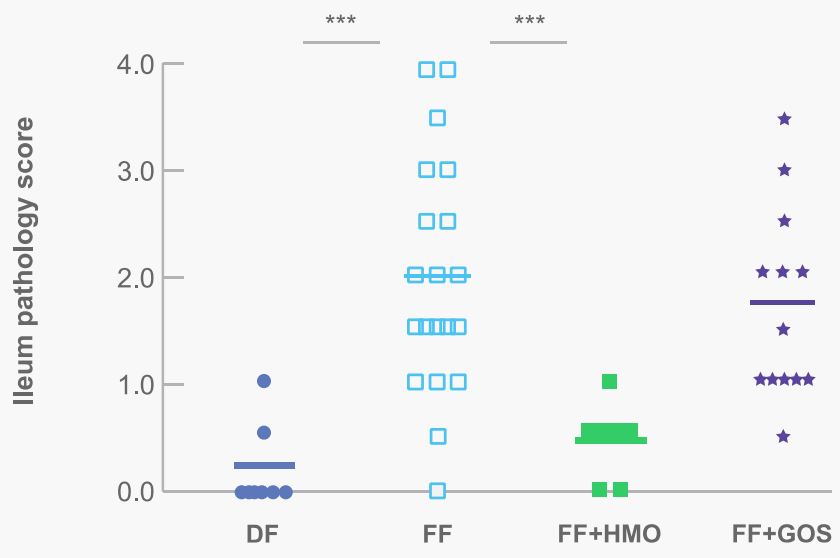
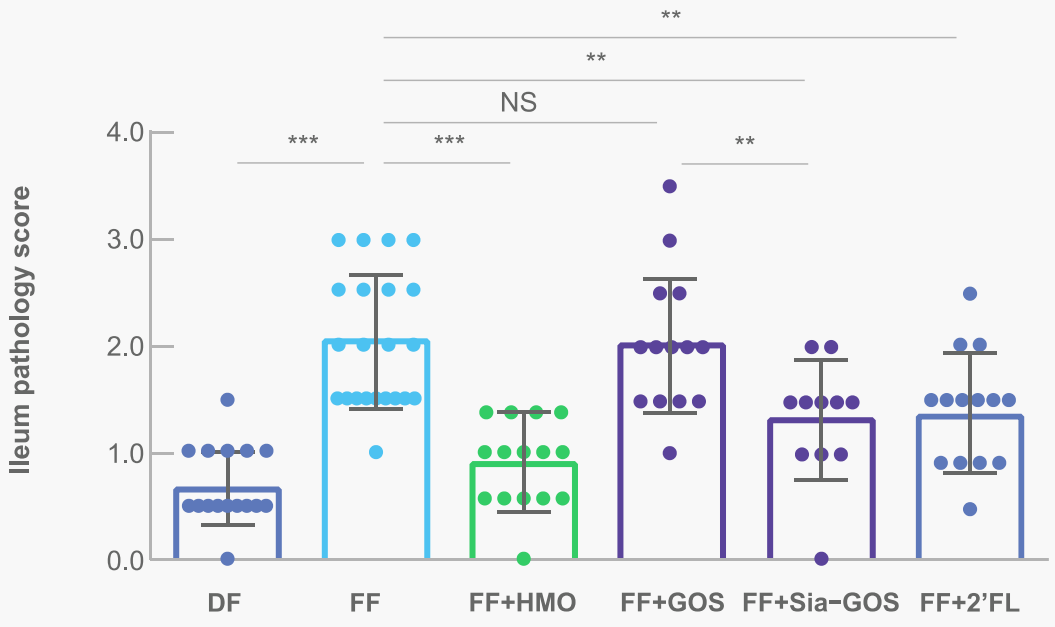
HMOs may play a role in mitigating the effects of NEC[1,2*]

Pooled HMO[1]



Pooled HMO, 2'FL[2]



*** p<0.001, **p<0.01

2'FL, 2-fucosyllactose; DF, dam fed; FF, formula fed; GOS, galacto-oligosaccharide; HMO, human milk oligosaccharide; NEC, necrotising enterocolitis; Sia, sialylated.

1. Jantscher-Krenn E et al., 2012; 2. Autran CA et al., 2016. * Animal model.

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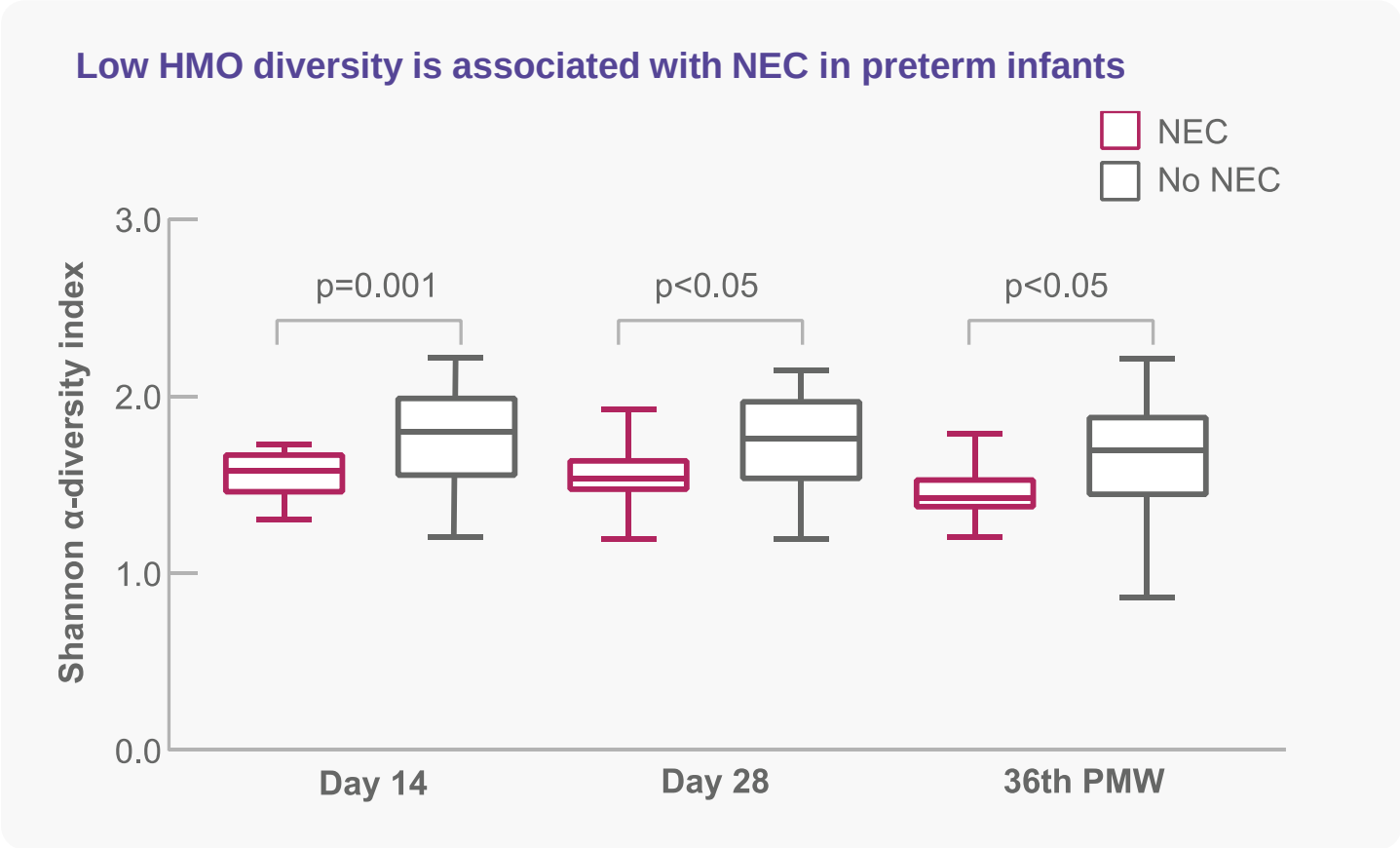
HMOs may play a role in mitigating the effects of NEC[1*]



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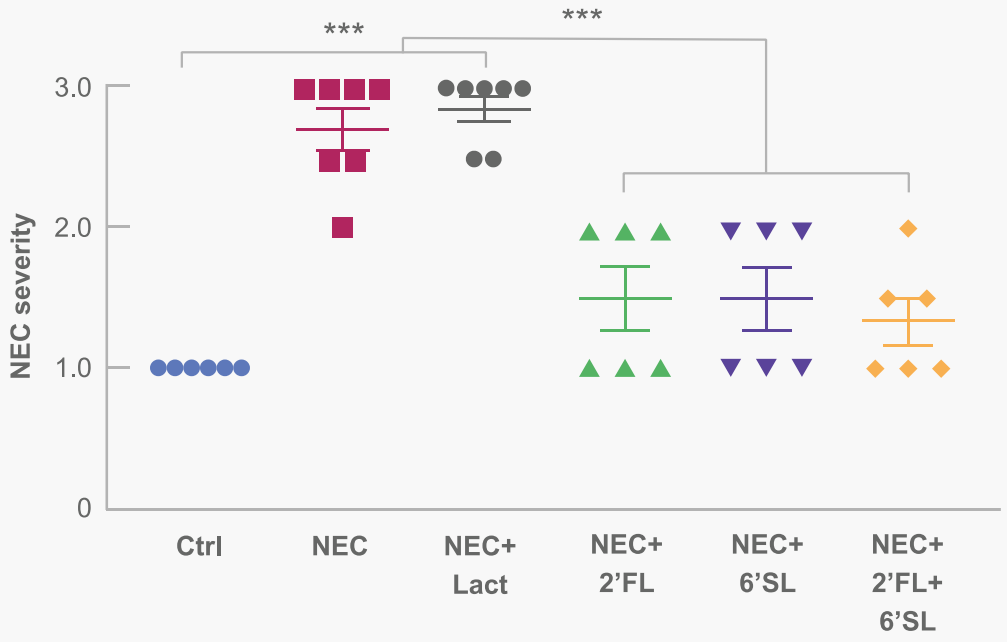
HMO, human milk oligosaccharide; NEC, necrotising enterocolitis; PMW, post menstrual week

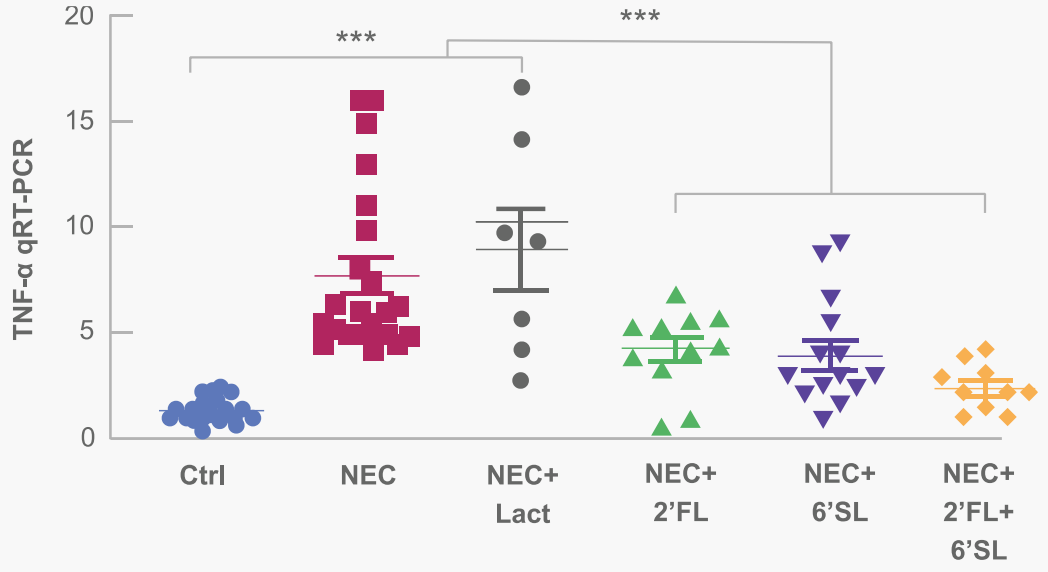
1. Wejryd E et al., 2018. *Observational study.

HMOs may influence NEC severity by reducing TNF-α and TLR4 mediated signaling[1*]

2’FL and 6’SL reduce NEC severity



2’FL and 6’SL reduce levels of proinflammatory TNF-α alpha



*** p<0.001

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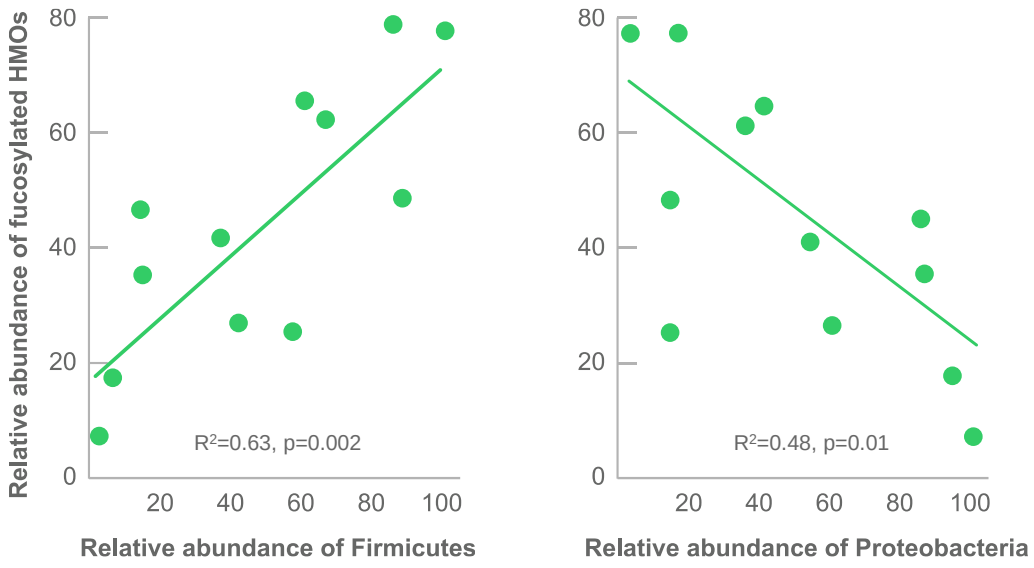
2’FL, 2-fucosyllactose; 6’SL, 6-sialyllactose; HMO, human milk oligosaccharide; Lac, lactose; NEC, necrotising enterocolitis; qRT-PCR, quantitative real-time polymerase chain reaction; TLR4, toll-like receptor 4; TNF-α, tumor necrosis factor alpha.

1. Sodhi CP et al., 2020. *Animal studies in combination with experimental studies

Fucosylated HMOs may reduce gut microbiota dysbiosis in premature infants[1*]

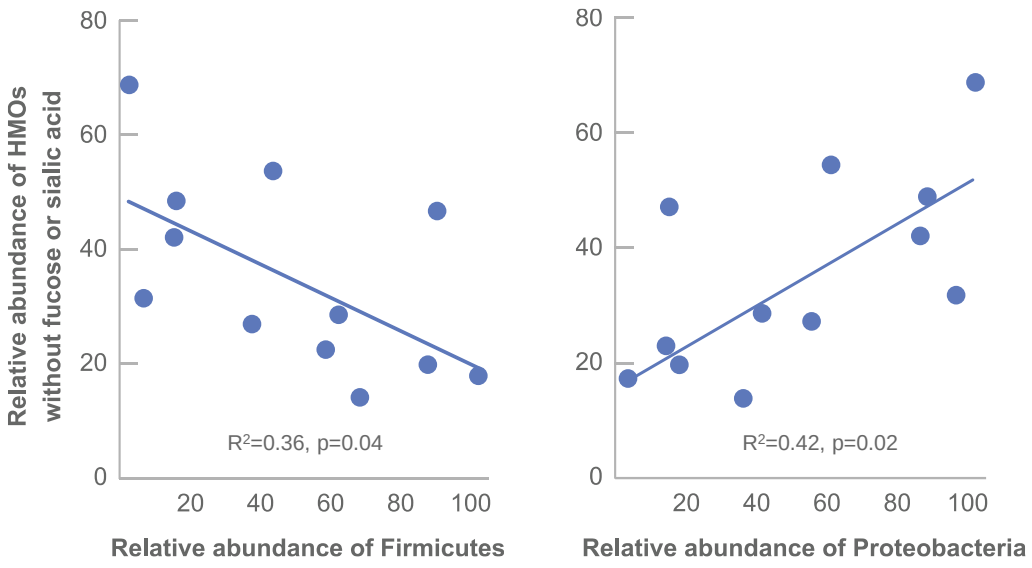
Fucosylated HMOs



Higher levels of fucosylated HMOs are associated with:

- Higher levels of commensal Firmicutes
- Lower levels of primarily harmful Proteobacteria

Non-Fucosylated/ non-sialylated HMOs



Higher levels of non-fucosylated or non-sialylated HMOs are associated with:

- Lower levels of commensal Firmicutes
- Lower levels of primarily harmful Proteobacteria

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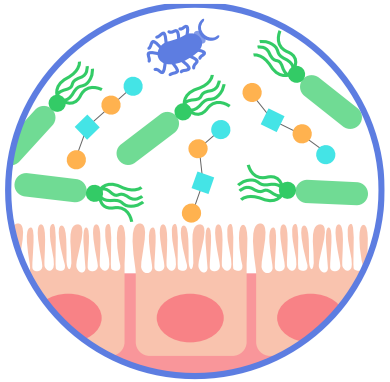
HMO, human milk oligosaccharide.

1. Underwood MA et al., 2013. *Observational study, n=12.

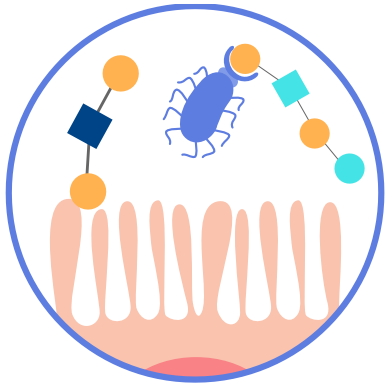
HMOs help the immune defences in 4 main ways[1-5]

Promote the growth of beneficial bacteria



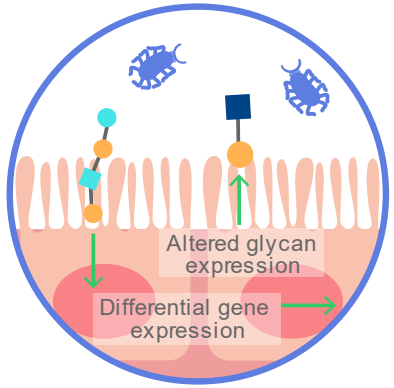
+HMOs

Prevent pathogen adhesion in the gut



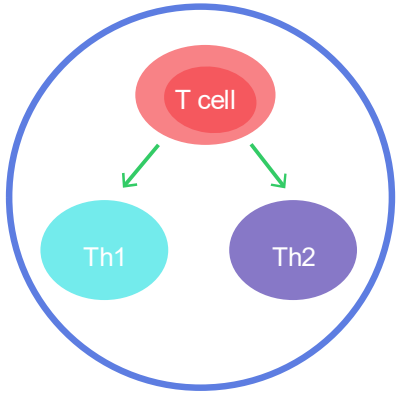
+HMOs

Strengthen gut barrier function



+HMOs

Educate the developing immune system



+HMOs

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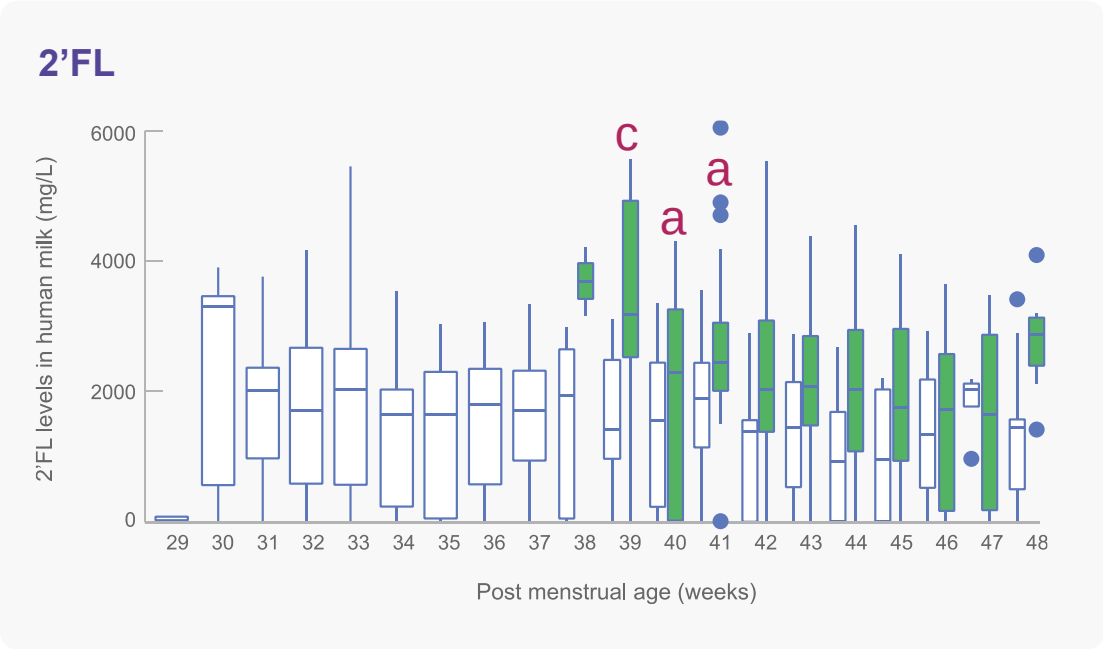
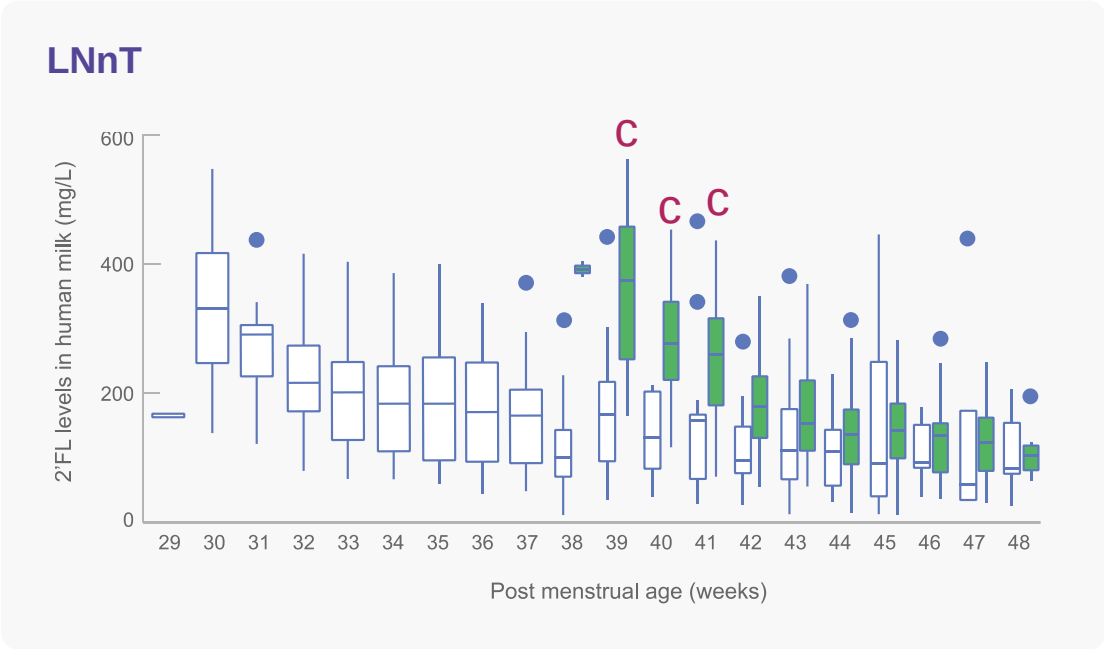
HMO, human milk oligosaccharide.

1. Underwood MA et al., 2013; 2. Zivkovic AM et al., 2011; 3. Zivkokiv AM et al., 2013; 4. Bode L, 2015; 5. Donovan SM et al., 2016.

Most levels of HMOs are lower in preterm milk compared to term milk, at equivalent developmental status[1]*

The largest differences are seen around 40 postmenstrual weeks





Term milk
Preterm milk

a, p<0.05
c, p<0.0005

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2'FL, 2-fucosyllactose; HMO, human milk oligosaccharide; LNnT, Lacto-N-neotetraose.

1. Austin S et al., 2019. *Observational study

Preterm infants may have insufficient HMO intake

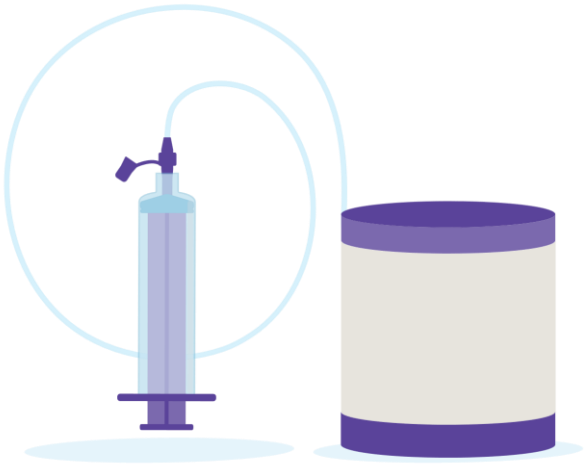
Mother's own milk

- Levels of HMOs may be reduced[1]

Donor milk

- Pasteurisation and sterilisation reduce levels of HMOs in donor milk[2,3]

Alternative feeds



- Do not currently contain HMOs

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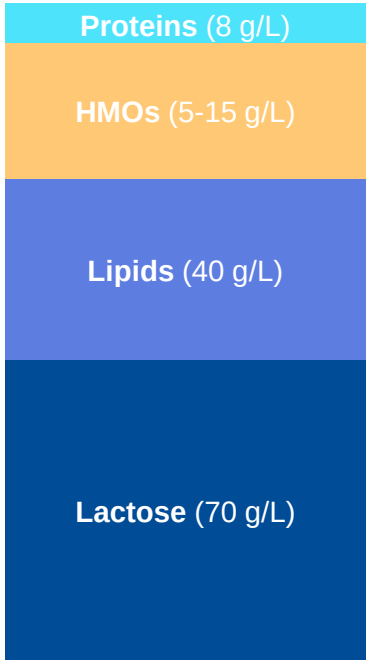
HMO, human milk oligosaccharide.

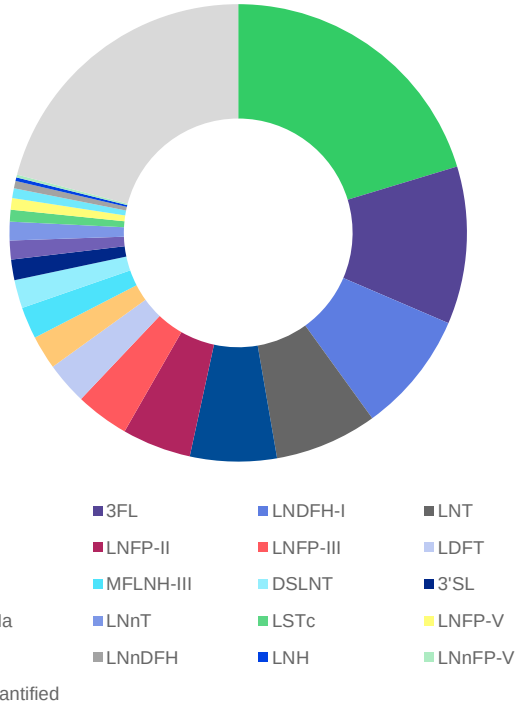
1. Austin S et al., 2019; 2. Marx C et al., 2014; 3. Meredith-Dennis L et al., 2018.

HMOs are the 3rd largest solid component in human milk

Solid components[1-3]



> 200 HMOs identified
20-30 HMOs quantified[4-11]



Three main HMO categories based on their building blocks[4-11]

- **Non-fucosylated neutral (core) HMOs**
  e.g., LNnT, LNT
- **Fucosylated neutral HMOs**
  e.g., 2'FL, DFL
- **Sialylated acidic HMOs**
  e.g., 6'SL, 3'SL

DFL, difucosyllactose (also named as LDFT, lactodifucotetraose); DFLNHa, difucosyllacto-N-hexaose a; DSNLT, disialyllacto-N-tetraose; FL, fucosyllactose; HMO, human milk oligosaccharide; LNDFH, lacto-N-difucohexaose; LNFP, lacto-N-fucopentaose; LNT, lacto-N-tetraose; LNnT, Lacto-N-neotetraose; LST, sialyl-lacto-N-tetraose; MFLNH, monofucosyllacto-N-hexaose; SL, sialyllactose.

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1. Zivkovic AM et al., 2011; 2. Zivkokiv AM et al., 2013; 3. Bode L, 2012; 4. Hennet T et al., 2014; 5. Bode L, 2015; 6. Thurl S et al., 2017; 7. Austin S et al., 2016; 8. Sprenger N et al., 2017; 9. Samuel TM et al.,2019; 10. Austin S et al., 2019; 11. Lefebvre G et al., 2020.

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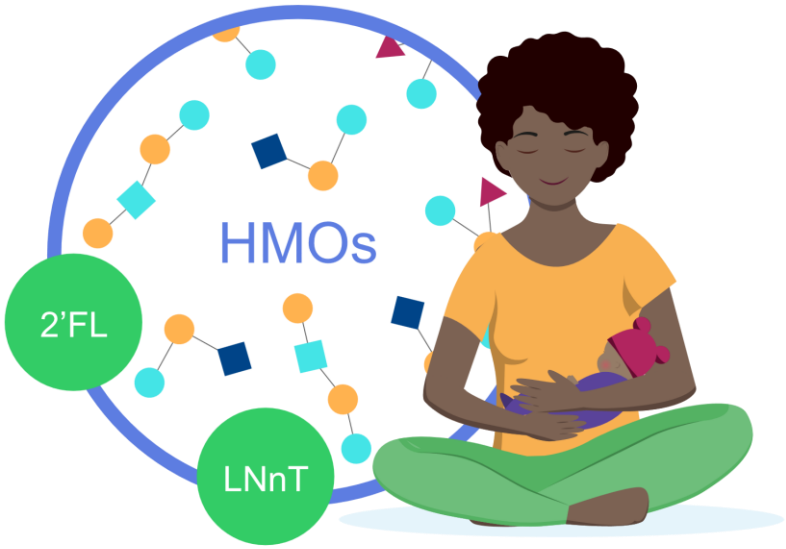
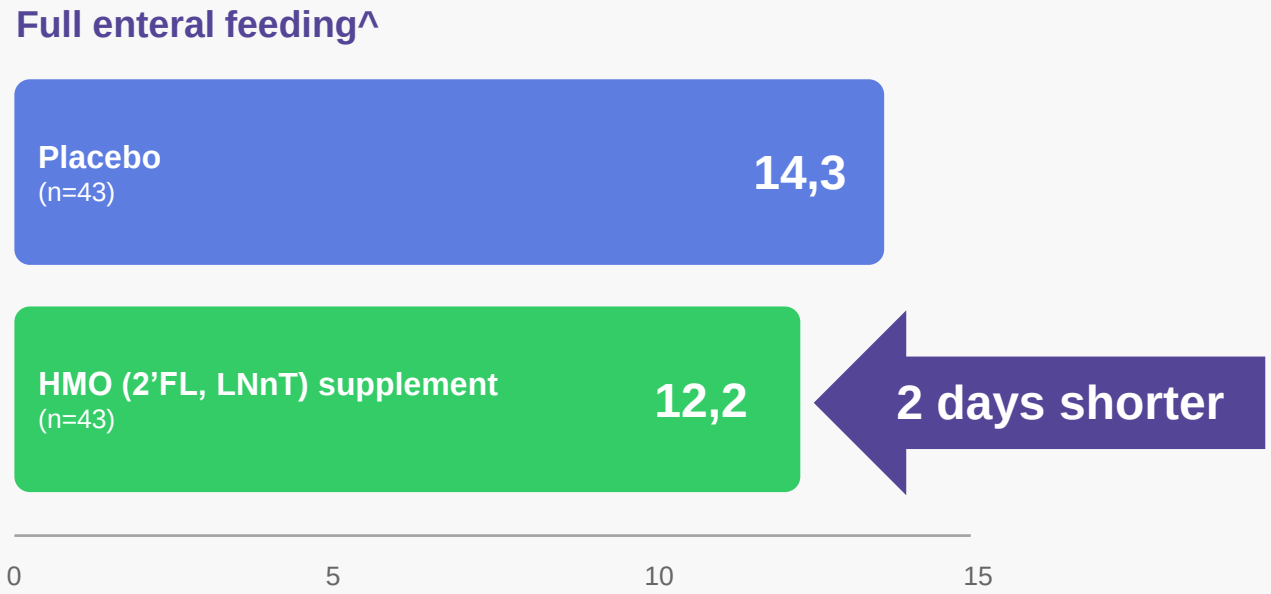
Clinical Evidence of a HMO Supplement (2’FL, LNnT) in Preterm Infants

HMO, Human milk oligosaccharide.

With HMO supplement (2’FL and LNnT), the mean time to reach full enteral feeding was 2 days shorter[1*]



Faster progression to FEF may help reduce the risk of growth restriction and exacerbated gut immaturity[1]



^Time to reach full enteral feeding is a reliable indicator of feeding tolerance

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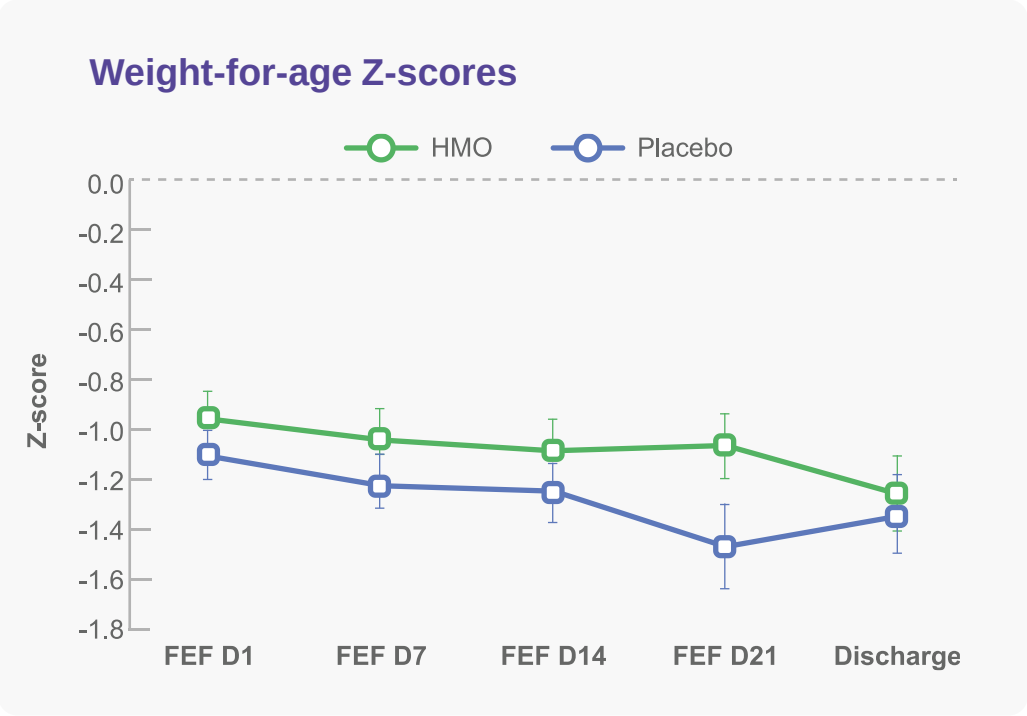
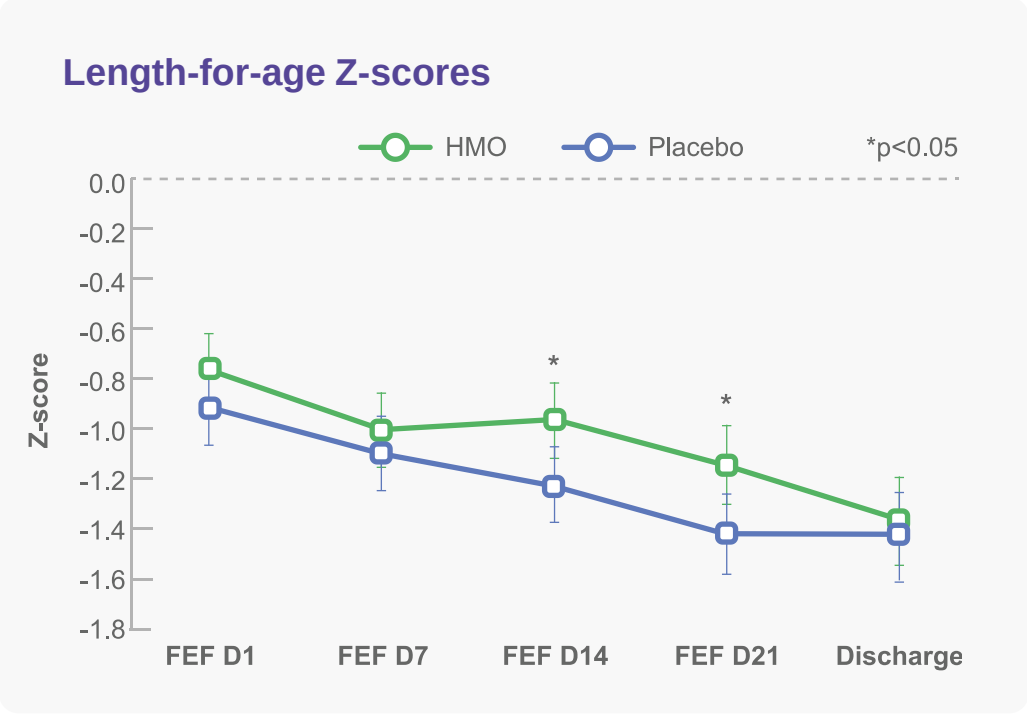
2’FL, 2-fucosyllactose; HMO, human milk oligosaccharide; LNnT, Lacto-N-neotetraose.

1. Hascoët JM et al., Abstract at WCPGHAN 2021. * Randomised, clinical trial

HMO supplement (2'FL and LNnT) supports adequate growth, in line with preterm growth standards^

HMO supplement supports early postnatal growth, which may have a positive impact on long-term growth and developmental outcomes[1*]





^Fenton growth standards for preterm infants

2'FL, 2-fucosyllactose; FEF, full enteral feeding; HMO, human milk oligosaccharide; LNnT, Lacto-N-neotetraose

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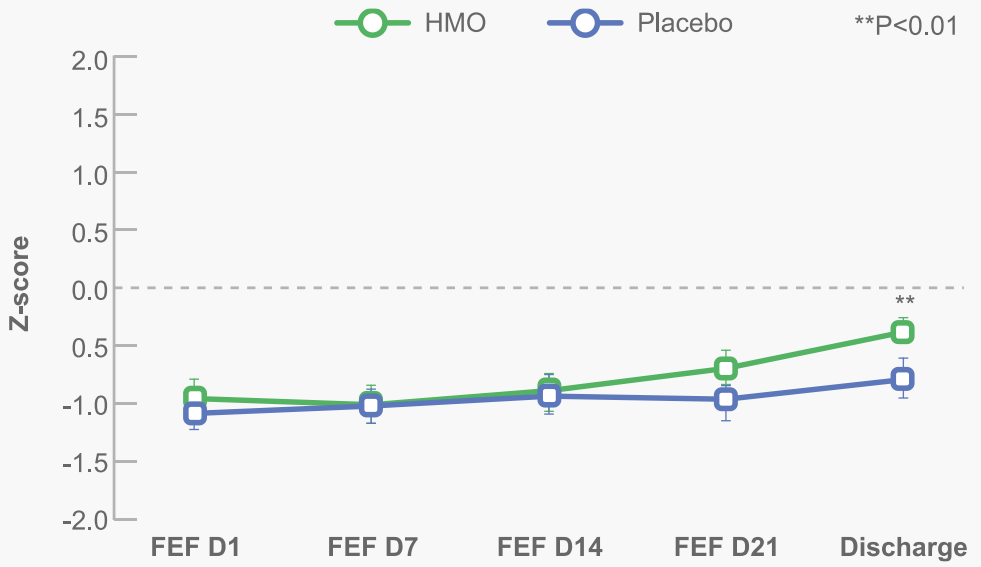

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HMO supplement (2'FL and LNnT) supports adequate gastrointestinal tolerance

Gastrointestinal tolerance[1]*	HMO Range of means	Placebo Range of means
Daily gastric residual volume (in mL/kg/day)	0.05 - 3.06	0.07 - 3.29
Daily stool frequency	3.09 - 4.17	3.11 - 4.20
Stool consistency (1=watery, 2=runny, 3=mushy soft, 4=formed, 5=hard)	2.97 - 3.11	2.91 - 3.19
Difference from stool consistency of 3 (mushy soft)	0.13 - 0.33	0.16 - 0.50

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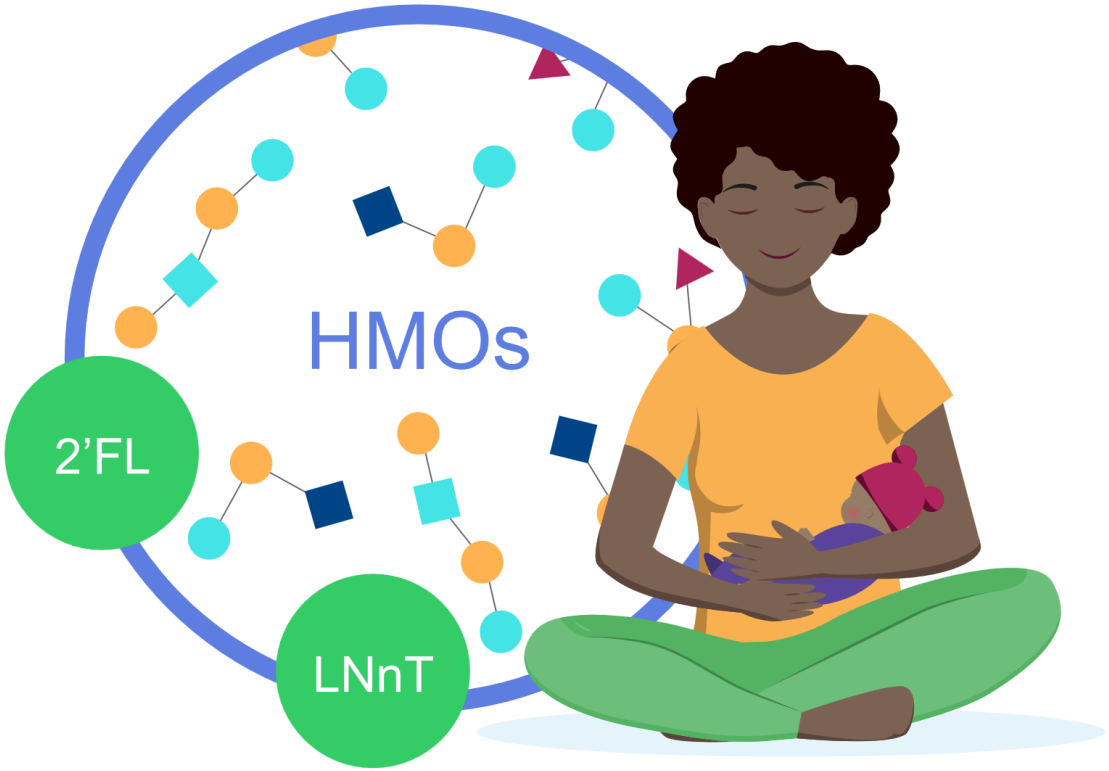
2'FL, 2-fucosyllactose; FEF, full enteral feeding; HMO, human milk oligosaccharide; LNnT, Lacto-N-neotetraose

1. Hascoët JM et al., Abstract at WCPGHAN 2021. * Randomised, clinical trial

Takeaway message



- Preterm infants are a vulnerable group of neonates which may have insufficient HMO intake.
- Supplementing the HMO 2’FL and LNnT promotes early full enteral feeding versus placebo.
- Faster progression to full enteral feeding may help reduce the risk of growth restriction and exacerbated gut immunity.
- HMO supplement (2’FL and LNnT) supports early postnatal growth, which may have a positive impact on long-term growth and developmental outcomes.

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2’FL, 2-fucosyllactose; FEF, full enteral feeding; HMO, human milk oligosaccharide; LNnT, Lacto-N-neotetraose.